

Hockey Goalie Stats

Main App User Manual

Version 5.78 manual edition
Comprehensive feature and workflow guide

Hockey Goalie Stats - Data Driven Development

Prepared for coaches, analysts and goalkeeper development staff

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1. Purpose and Core Concepts

Hockey Goalie Stats is a local-first goalkeeper analysis system. It stores goalkeeper profiles, team classifications, matches, shot records, shot locations, period order, coaching ratings, video-review metadata, reports and backups. The app is designed for live recording, companion-recorder imports, and detailed post-match video analysis.

The same shot model is used throughout the app. A shot recorded live, imported from the Recorder, or captured in Video Stats Review contributes to the same statistics, heat maps and reports.

The three record outcomes

Outcome	Meaning	Statistical treatment
Save	A shot on target is stopped by the goalkeeper.	Counts in Save Rate and Defence Rate.
Goal	A shot results in a goal against.	Counts in Save Rate and Defence Rate.
Angle Closed Off	The goalkeeper action prevents the attempt becoming an on-target shot.	Excluded from Save Rate; counts positively in Defence Rate.

Rate definitions

- Save Rate = Saves divided by Saves plus Goals. Angle Closed Off records are excluded.
- Defence Rate = Saves plus Angle Closed Off divided by all shot records.
- N/C means the selected records do not contain enough applicable events for that calculation.
- Opponent goals are calculated from recorded Goal shots. Confirm the shot list if the score does not agree.

> Good practice: Consistent definitions matter more than recording every possible detail. Agree on recording rules before a season starts and apply them consistently.

2. Installation, Storage and Navigation

Installing in Chrome

1. Open the hosted Hockey Goalie Stats address in Chrome.
2. Use the browser installation icon or menu option labelled Install app.
3. Confirm installation. The app then opens in its own window while continuing to use the browser storage on that device.

Local-first storage

- Goalie, match and shot data is saved in the browser on the current device.
- Large videos are not copied into the normal database backup. Keep them in their original folders or on an external drive.
- Browser data can be lost if site storage is cleared, the browser profile is removed, or the device fails. Use the Backup tab regularly.
- The app works offline for local features once installed and cached. Benchmark upload/download requires an internet connection.

Sidebar navigation

- The desktop sidebar remains in place while the selected page scrolls independently.
- Use Collapse Menu to reduce the sidebar to abbreviated buttons.
- On smaller screens, open the slide-out menu from the header.
- Video Stats Review is opened from a match rather than from the sidebar. Its Return button takes you back to the tab from which it was opened.

3. Home Tab

Home is the quick health check for the local database. It shows the current totals for goalkeeper profiles, matches, shot records and the installed app version.

Use these totals after a restore or import to confirm that the expected information is present. The totals are not filtered by the currently selected goalkeeper.

4. Goalies Tab

Creating and selecting a goalie

1. Select New Goalie.
2. Enter the goalkeeper name, date of birth and gender.
3. Review the automatically calculated age group.
4. Add at least one Current Team Profile.
5. Select Save Goalie.

Current Team Profiles

Each current team stores a team name, age group, team level and team tier. Levels include School, Club, Provincial, National and Other. Tiers use A / 1st, B / 2nd, C / 3rd, D / 4th, Other or Not specified.

The chosen home team profile supplies the home age group, level and tier when a match is created. These home values are historical match data and are not edited directly in Match or Match Editor.

- Add Team creates another team profile.
- Apply to Previous Matches deliberately replaces the saved home classifications in previous matches using that team name.
- Remove Team removes it from the current-team list but does not erase the team from historical matches or report filters.
- Goalie History lists saved matches for the selected goalie.

> Good practice: If a goalkeeper changes teams, keep the old matches. Remove the old team only from Current Team Profiles; historical reporting will retain it.

Deleting a goalie

Delete Selected Goalie permanently removes the goalkeeper and their stored matches. The app displays a randomly generated six-digit code that must be typed exactly. Make a backup first.

5. Match Tab - Match Details

Starting a match

1. Choose the goalkeeper, date and home team.
2. Enter the opponent. The app builds the match name automatically and numbers repeated same-day fixtures when necessary.
3. Choose the number of periods: 1, 2, 4 or Custom.
4. Choose Match Type and Goalkeeper Participation. Enter Minutes Played for a partial match.
5. Review the opposition age group, level and tier.
6. Select Start / Save Match Details.

Classifications and automatic opposition defaults

Home classification is taken from the goalkeeper team profile. When starting a match, the opposition classification initially follows the home classification. Once any opposition classification is changed manually, later home changes do not overwrite that opposition choice.

Periods

- Choose 1, 2 or 4 when the competition format is known.
- Choose Custom for another format. The effective number of periods is calculated from the recorded End of Period markers.
- Period markers determine how shots are assigned to periods and how multi-period reports are built.

6. Match Tab - Recording Shots

Shot fields

Field	Options and use
Shot Outcome	Save, Goal, or Angle Closed Off.
Shot Situation	Normal Game Play, Penalty Corner, Penalty Stroke, or 8 Second 1v1.
Shot Type	1st Shot, Rebound, In Game 1v1, Own Goal, or Other.
Out Numbered	Not Out Numbered, 2 vs 1, 3 vs 1, or 4+ vs 1.
Rebound Result	No rebound, Safe, or Dangerous.
Notes	Optional context for later review.

Saving with heat maps

- For Save and Goal, place both a D location and a Goal Box location.
- For Angle Closed Off, place the D location only.
- Penalty Stroke uses the fixed penalty spot where the workflow applies it.
- Use Mirror goal-box tap when the view is recorded from behind the goal and the displayed orientation needs reversing.

Saving without heat maps

Save Shot Without Heat Map is intended for rapid live capture. The complete shot detail is saved immediately and its missing placements can be added later in Match Editor. Clear Shot Options resets the current entry controls without deleting saved records.

> Good practice: During fast play, prioritise outcome and shot classification. Missing placements can be corrected later; a missed event is harder to reconstruct.

7. Match Tab - Periods, Outcomes and Ratings

End of period markers

Use End of Period at the moment a period ends. The marker is added to the same ordered timeline as shots. It defines which shots belong to each period and is essential for accurate period and half reports.

Match outcome

- Enter Home Team Goals.
- Opponent Goals is calculated from Goal records.
- Complete the 1-to-5 coaching ratings for Positioning, Communication, Rebound Control and Decision Making.
- Save Match Outcome & Prepare Next Match finalises the match and clears the live page for the next match.
- Clear Match Progress & Start Again removes the active draft, including its shots and Video Stats Review metadata, after six-digit confirmation.

Shootouts

When a match is recorded as a draw, the shootout workflow can store each round, the opposition attempt result, goalkeeper save method and note, plus sudden-death rounds. Shootout information is reported separately from normal match shot statistics.

8. Match Editor Tab

Match Editor is the correction and completion workspace for saved matches. Choose a goalie and match first.

Edit / Delete Match

- Edit date, opponent, score, period option, status, match type, participation and minutes played.
- Home classification remains tied to the saved team information and is read-only.
- Opposition classification can be corrected independently.
- Set Draft or Finalised explicitly when repairing an older match.
- Deleting a match requires the six-digit confirmation code.

Shots and period markers in recorded order

- Add a missing shot or End of Period marker.
- Move shots and markers up or down to restore the correct order.
- Edit shot outcome, situation, type, outnumbered state, rebound result and notes.
- Delete incorrect shots. End of Period markers can be deleted directly and should then be rechecked against the match structure.

Game Heat Map Editor

The editor shows how many recorded shots still need D or Goal Box placement. Choose Place Save, Place Goal or Place Angle Closed Off, then select the map position. Remove Placement removes the chosen dot without deleting the shot. Edit Shot Placement targets a specific shot.

9. Stats Tab

Stats combines the selected goalkeeper records using filters for home team, opponent and reporting period. Period choices are Match, Year, Last 5 Games and All Matches. Select Update Stats after changing filters.

- Overall cards include Save Rate, Defence Rate, games and outcome totals where applicable.
- Breakdowns use shot situation, shot type, outnumbered state, rebound result and other recorded dimensions.
- Include Benchmarks becomes available only when benchmarking is activated and comparison data has been downloaded.
- Benchmark figures shown beside stats use a matching anonymous cohort; unavailable comparisons remain clearly identified.

10. Heat Maps Tab

Choose a goalkeeper and view: one Match, Last 5 Games, or Year. Select Update Heat Map Editor to refresh the visualisation.

Colour	Meaning
Green	Save
Red	Goal
Yellow	Angle Closed Off

- The D Heat Map shows the origin or defensive engagement location.
- The Goal Box Heat Map shows where an on-target Save or Goal was directed.
- Use Match Editor when a dot is missing or placed incorrectly.

11. Video Stats Review - Opening and Layout

Open Video Stats Review from Match or Match Editor. The selected match remains active, and Return to Match / Return to Match Editor follows the route used to open it.

The workspace includes the video stage, playback and framing controls, a unified video timeline, collapsible Video Angles panels, the shot recorder, heat maps, match timeline and clip export tools.

Reset behaviour

Starting or finalising a match clears the Video Stats Review page for the next match. Reset Video Stats Review clears the current review workspace when a complete restart is required.

12. Video Stats Review - Video Libraries, Angles and Camera Runs

Recommended external-drive structure

Match folder / Angle folder / optional numbered Camera Run folder / video clips

The match folder should use the generated Match Folder Name. Each angle folder name becomes the angle name. Optional folders named 1, 2, 3, 4 and so on become Camera Runs for that angle. Clips inside each run are sorted into the playback sequence.

Example folder	Purpose
2026-09-04 - Home VS Opponent	Match folder
Angle 1	Angle name
Angle 1 / 1	Camera Run 1
Angle 1 / 2	Camera Run 2
Behind Goal	A second angle without numbered runs

External Video Library controls

- Copy Folder Name copies the exact match folder name for easy renaming.
- Choose Video Library Folder links the main external-drive library root.
- Use Existing Match Folder links a folder already arranged for the selected match.
- Import Match Folder to Library copies source videos into the selected external library without deleting originals.
- Allow Video Access restores Chrome folder permission after a restart when required.
- Batch Add Videos loads a structured match folder directly.

Good practice: Chrome or Edge is required for full folder access. Keep the drive and folder names stable. The app stores permission handles and metadata, not a second copy of the videos in its database.

Video Angles panels

- Switch Angles changes playback to a named loaded angle.
- Add and Batch Videos adds individual clips or a structured folder.
- External Video Library manages persistent folder access.
- Loaded Video Angles manages clips, order and angle settings.
- Each panel can collapse to its heading to reduce clutter; its state is remembered.

13. Video Stats Review - Playback, Framing and Sync

Playback controls

- Play/Pause, -5 seconds, frame backward and frame forward support detailed review.
- Playback speed options include slow motion and accelerated scanning.
- The long seek bar uses the current Camera Run duration and resets correctly when runs are changed.
- Only one angle plays at a time. Switching angles keeps the match-time relationship.

Framing

Use Zoom -, Zoom + and drag/pan the image to reframe portrait or wide source footage. Reset Framing restores the default view. Framing is stored separately for each angle and each Camera Run and is applied to exported clips.

Camera Run playback state

Each run retains its own clip list, active clip, position, sync information and framing. Changing Camera Run must load that run timeline rather than leaving the previous run at its end position.

Manual and waveform sync

- New Sync Point creates a shared match reference point.
- Sync Current Clip links the current clip moment to the selected sync point.
- Clear Clip Sync removes that clip link.
- Delete Unused Sync Point removes a reference no longer used.
- Show Audio Waveforms builds progressive waveforms for the selected Camera Run.
- Suggest Auto Syncing uses audio similarities to propose alignment. Review suggestions before relying on them.
- Manual sync remains available even when auto sync or waveforms are unavailable.

14. Video Stats Review - Recording and Timeline

1. Seek to the shot moment using the active angle and Camera Run.
2. Choose outcome, situation, shot type, outnumbered state and rebound result.
3. Place the required D and Goal Box locations.
4. Select Save Video-Linked Shot.
5. Use End Period at the actual end of each period.
6. Review the Match Timeline. Selecting a video-linked shot returns playback to its saved moment.

A video-linked shot stores the same statistical fields as a live shot plus its match-time and video references. Period markers determine the period shown in clip lists and overlays; gameplay clock time is intentionally omitted from the overlay because it may not match the video precisely.

15. Video Stats Review - Clip Export and Second Screen

Clip export

- Set Clip In and Set Clip Out create a manual range for Export Selected Clip.
- Before shot and After shot define the automatic window around a selected video-linked shot.
- Export Current Shot Clip exports the selected shot for the active angle.
- Export Every Shot - Current Angle creates one MP4 per video-linked shot for that angle.
- Export Every Shot - Every Angle creates all available shot-angle clips.
- Include Shot Details Overlay adds the match and shot classification overlay with the complete logo and without gameplay time.
- Exports preserve the selected run framing and include source audio when the browser and source codec permit it.
- Chrome may request permission for multiple downloads during a batch. Choose Allow.

Second-screen player

Pop Out Player opens playback in a separate browser window for a two-screen setup. It mirrors active footage and provides angle, Camera Run, seek, speed, volume, framing and full-screen controls. Keep the main app open for recording and use the second window for video display.

16. Reports Tab

Choose the goalkeeper and Report Type, complete the report-specific filter, then select Generate Report. Print Report opens the browser print workflow; choose Save as PDF when a digital copy is required.

Report	Use
Individual Match Report	One selected match with match-specific statistics and recorded detail.
5 Game Report	The goalkeeper's five most recent matches.
Year Report	All qualifying records in the selected calendar year.
Goalie Development Report	Long-term match trend with recent five versus previous five rates.
Opponent Analysis Report	Performance against one selected opponent.
Team / Home Team Report	Performance while representing one selected home team.
Shot Situation Report	Year breakdown by Normal Game Play, Penalty Corner, Penalty Stroke and 8 Second 1v1.
Period Report	Period/half analysis using End of Period markers. In four-period games, periods 1-2 form First Half and 3-4 form Second Half.
Training Focus Report	Ranks focus areas using performance weakness, event volume, goals and risk; small event counts are marked limited.
Data Driven Development Report	Comprehensive selected-range report with profile, matches, trends, breakdowns, heat maps, shootouts, focus and timelines.
Benchmark Report	Compares Full Year or Last 5 Matches against matching anonymous cohorts by age, gender, level and tier.

Good practice: Check the Matches Included section before distributing a report. A technically correct report can still be misleading if the wrong year, team, opponent or range was selected.

17. Anonymous Benchmarking Tab

Consent and access

- Benchmarking is optional. Select Opt In to participate or Opt Out to keep all normal local features without benchmark comparisons.
- If opted in, all eligible matches are included; the user cannot selectively upload only favourable matches.
- Benchmark figures unlock after a successful eligible upload.
- Finalising an eligible match attempts an automatic update when online. Upload Data & Refresh Benchmarks forces a manual upload and cohort refresh.
- Repeated uploads do not duplicate unchanged matches; stable anonymous IDs allow central records to be updated.

What is uploaded

The benchmark payload uses an anonymous goalkeeper identifier and anonymous match identifier. It keeps the goalkeeper gender and age group at the match, home and opposition level/tier/age classifications, match type, participation, minutes, period count, aggregate rates and anonymous shot classifications/locations. It does not send the goalkeeper name or team names.

Eligibility

- The match must be Finalised.
- At least one shot must be recorded.
- Goalkeeper and opposition classifications, match type and participation must be complete.
- Partial match participation requires Minutes Played.

Fairness and limited data

- Save Rate and Defence Rate benchmark calculations use each goalkeeper's average across qualifying matches so one frequent uploader does not dominate the cohort.
- Benchmark heat maps include every qualifying anonymous shot.
- Small cohorts remain visible but are labelled Limited Data when there are fewer than 5 goalkeepers, 15 matches or 100 shots.
- Compare like with like by selecting appropriate age group, gender, team level and tier.

18. Backup Tab

Complete database backup

- Download JSON Backup saves all local app data.
- Restore Backup replaces the local database with the chosen complete backup.
- Clear All Local Data permanently removes the browser database and should only be used after a verified backup.

Single Goalie Transfer

- Export Selected Goalie includes the selected profile, team records, matches, shots, timelines and Video Stats Review metadata.
- Import & Merge Goalie merges by stable IDs instead of replacing other local goalies.
- Newer versions overwrite older versions; older backups cannot overwrite later edits.
- Incoming team or match deletions are listed and require approval. Declining deletion still permits other newer additions and updates to merge.
- Video files themselves are not embedded. Reconnect them from the receiving coach's drive.

Recorder import and master data

- Import Match From Goalie Stats Recorder loads the Recorder JSON into the currently selected goalie, or creates an imported goalie when none exists.
- Export Goalies & Teams creates a lighter master-data file containing profile/team setup rather than the full shot database.

19. Resources Tab

Resources packages the Data Driven Development theory document, current FIH Rules of Hockey, the Main App User Manual and the Recorder User Manual. Open displays a document in the browser; Download saves a copy. Manuals should be refreshed whenever the app or Recorder workflow materially changes.

20. Workflow A - Record Directly in Match at the Venue

1. Before the match, confirm the goalie profile and team classifications.
2. Open Match and create the fixture with the correct period format, type and participation.
3. During play, record each event immediately. Use Save Shot Without Heat Map when speed matters; otherwise place both heat-map points.
4. Record End of Period exactly when each period finishes.
5. At full time, confirm Goal records against the scoreboard, enter home goals and coaching ratings, then finalise the match.
6. After the match, use Match Editor to add missing placements, correct shot order and verify classifications.
7. Review Stats and Heat Maps, then create the required report and make a backup.

Good practice: Best for a coach with a clear view and enough time to classify events. A second person can improve accuracy by recording while the coach observes play.

21. Workflow B - Record with the Companion Recorder

1. Before the match, prepare the goalkeeper/team and fixture information in the Recorder.
2. Record shots and End of Period markers live. Prioritise correct order and outcome; add precise placements where practical.
3. Review the Recorder timeline at half time and after the match.
4. Export the completed match JSON from the Recorder and retain a copy.
5. In the main app, select the intended goalkeeper, open Backup, choose the Recorder Match File and import it.
6. Open Match Editor to confirm the match, classifications, order, score and missing heat-map locations.
7. Finalise if necessary, inspect Stats/Heat Maps and make a full main-app backup.

Good practice: Best when a phone or tablet is used pitch-side and the full analysis database is kept on another computer.

22. Workflow C - Film First and Record in Video Stats Review

1. Create the match before or after filming so Video Stats Review has a destination match.
2. Copy footage to an external drive using the match / angle / Camera Run folder structure.
3. Open Video Stats Review and connect the match folder. Grant Chrome video access if requested.
4. Check each angle and Camera Run, then set framing for each run.
5. Create sync points and sync clips manually, or use waveforms and auto-sync suggestions as an aid.
6. Work through the match, recording every shot and End of Period marker at the correct video moment.
7. Review the match timeline and correct any classifications or placements.
8. Export coaching clips with the desired before/after window and overlay.
9. Return to Match or Match Editor, complete the outcome/ratings, finalise, run reports and back up the database.

Good practice: Best for the most detailed analysis and for producing coaching clips. It takes longer but allows frame-by-frame decisions, multiple angles and accurate video-linked records.

23. Data Quality Checklist

- Correct goalie, date, home team and opponent.
- Correct age group, level and tier for both teams at the match date.
- Correct match type, participation and minutes played.
- Goal records agree with goals conceded.
- Shot order matches play order.
- End of Period markers are present and correctly positioned.
- Every Save/Goal has D and Goal Box placement where the chosen workflow requires it.
- Every Angle Closed Off has a D placement and no Goal Box placement.
- Rebound and outnumbered classifications are applied consistently.
- Match is Finalised only after review.

24. Troubleshooting

Problem	Action
Videos unavailable after restart	Reconnect the external drive and choose Allow Video Access or reselect the saved match folder.
Camera Run appears at the end	Change runs and allow the selected run to load; verify its clip files and duration metadata.
Waveforms are slow or fail	Generate only the required Camera Run, keep the tab active, and use manual sync if the source audio cannot be decoded.
Export framing is wrong	Set framing on the exact angle and Camera Run, then preview before exporting.
No export audio	Confirm the source contains audible audio and use a Chrome-supported source codec.
Stats do not match the score	Review Goal shots, own goals and shootout data separately in Match Editor.
Period labels are wrong	Correct the End of Period marker order before regenerating reports or overlays.
Benchmark unavailable	Confirm Opt In, eligibility, internet access, a successful upload and the existence of other matching cohort data.
Import may remove records	Read the deletion list. Approve only expected deletions; Cancel keeps them while importing other updates.

7-a-side match recording

Field Hockey Goalie Stats · Version 5.110 addendum

Choose the match format when creating or editing a field hockey match. This keeps 7-a-side records easy to find while the existing shot-recording workflow remains the same.

Set the format

- **New Match:** In Match Details, select **7-a-side** in Match Format before starting or saving the match. Matches default to 11-a-side.
- **Existing match:** Open Match Editor, choose the match, change Match Format if needed, then save the match details.
- **Periods:** Set the number of periods as appropriate for your competition. The existing period and event recording controls apply to both formats.

Review 7-a-side records

Use the Match Format selector in Stats or Reports to view All formats, 11-a-side, or 7-a-side. The match list includes a Format column so each game can be identified at a glance. Choosing All formats combines the records.

Benchmarks

Benchmark reports keep 7-a-side and 11-a-side matches in separate comparison groups. Older saved matches without a format are treated as 11-a-side, so you can edit any legacy match that was actually played as 7-a-side.

> **Age groups are calculated at 23:59 on 31 December of the previous year. A goalkeeper with a 1 January birthday remains eligible for the age group they were in at that cutoff.**

For competition-specific rules about squad size, field dimensions, match duration, or substitutions, follow the regulations for your event.